RIVER VALLEY HIGH SCHOOL
YEAR 6 PRELIMINARY EXAMINATION II

CANDIDATE NAME					
CENTRE NUMBER		INDEX NUMBER		CLASS	

H2 BIOLOGY **9648/02**

Paper 2 Core Paper **15 September 2014**

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Section A
Answer **all** questions in the spaces provided on the question paper.

Section B
Answer any **one** question on the answer paper provided.
Circle the question attempted on the cover page.

The use of an approved scientific calculator is expected, where appropriate.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together. The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	**/10**
2	**/ 12**
3	**/ 12**
4	**/ 12**
5	**/ 12**
6	**/ 10**
7	**/ 12**
Section B	
8 or 9*	**/ 20**
Total	

This Question Paper consists of **21** printed pages.

Section A (80 marks)

Answer **all** the questions in this section.

1. DNA polymerases carry out DNA replication during cell division, with tight coordination of leading and lagging strand synthesis. **Fig. 1.1** shows two representations of a bacterial DNA polymerase, DNA Pol III: a space-filling model and a ribbon diagram.

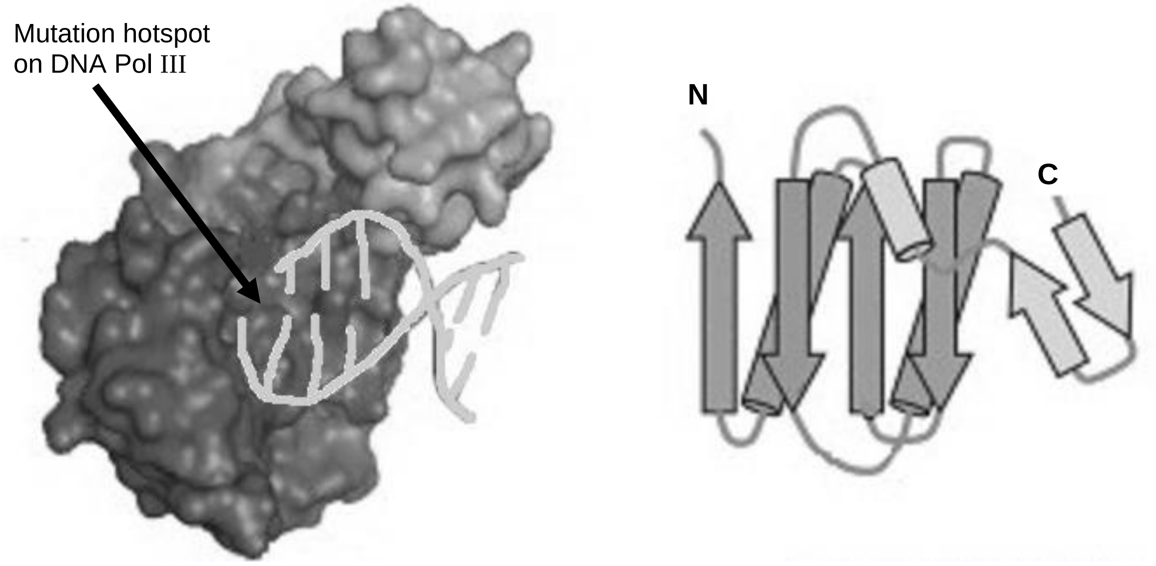

Image adapted from Cell, Vol 126, Issue 5

Fig. 1.1

(a) Describe the structure of DNA Pol III. [2]

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(b) Explain the significance to its activity of the structure of DNA Pol III. [2]

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1. **(c)** DNA Pol III is often used in laboratory for the synthesis of DNA under high temperature. Suggest one structural property that makes it a suitable molecule for such processes. [1]

(d) The arrow in **Fig. 1.1** shows a mutation hotspot on DNA Pol III where substitution mutation is common.

State two mutations that would not affect the activity of DNA Pol III. Explain your answer. [5]

[Total: 10]

2. The galactose operon (*gal* operon) controls the utilization of galactose in *Escherichia coli*. The operon consists of three genes (*gal*E, *gal*T and *gal*K) that code for enzymes involved in breaking down of galactose (Gal) into UDP-galactose (UDP-Gal) and glucose-1-phosphate (Glu-1-P), which can be used for cell wall synthesis and cellular metabolism.

Similar to the gene regulation of *lac* operon, the *gal* operon is also under dual control: negative control by the *gal* repressor, which is encoded by *gal*R gene and positive control by cAMP receptor protein (CRP), which is activated by cAMP in the absence of glucose.

Fig. 2.1 shows the structure of a *gal* operon and the biochemical pathway involving the enzymes encoded by the *gal* operon.

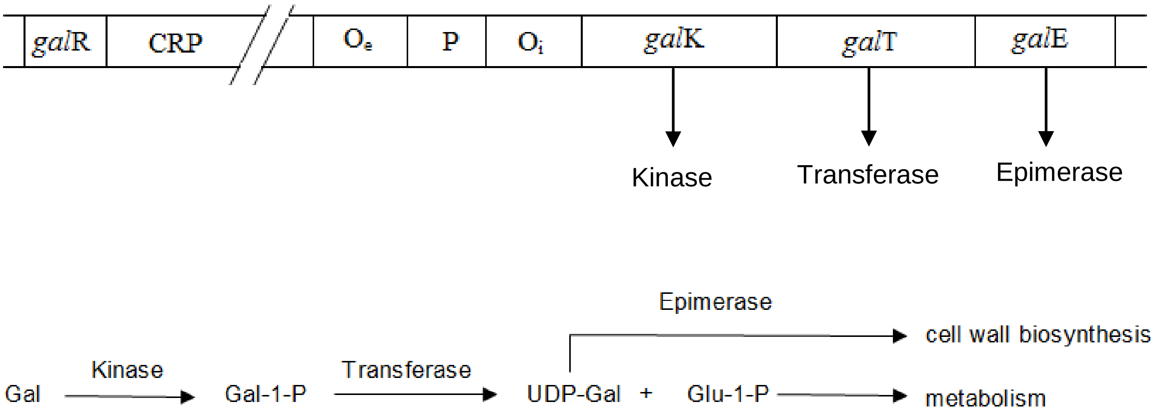

Image adapted from http://www.pearsonhighered.com & https://www.bio.cmu.edu

Fig. 2.1

(a) With reference to **Fig. 2.1**,

(i) describe how the *gal* operon is regulated when galactose is added to an *E.coli* suspension. [2]

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(ii) describe the change in regulation of the *gal* operon when glucose is added to the suspension in **(a)(i)**. [2]

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2. **(b)** In a mutated strain of *E.coli*, the bacteria can use galactose for metabolism but was unable to synthesise cell wall due to a mutation in the *gal* operon.

State and explain the location of the mutation. [3]

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λ bacteriophage is known to infect *E.coli* and can undergo both lytic and lysogenic cycles in the host cells. The bacteriophage inserts its own genome between the *gal* operon and *bio* operon.

Fig. 2.2 shows how the λ bacteriophage infects *E.coli*. However, an error occurred during the excision of phage DNA.

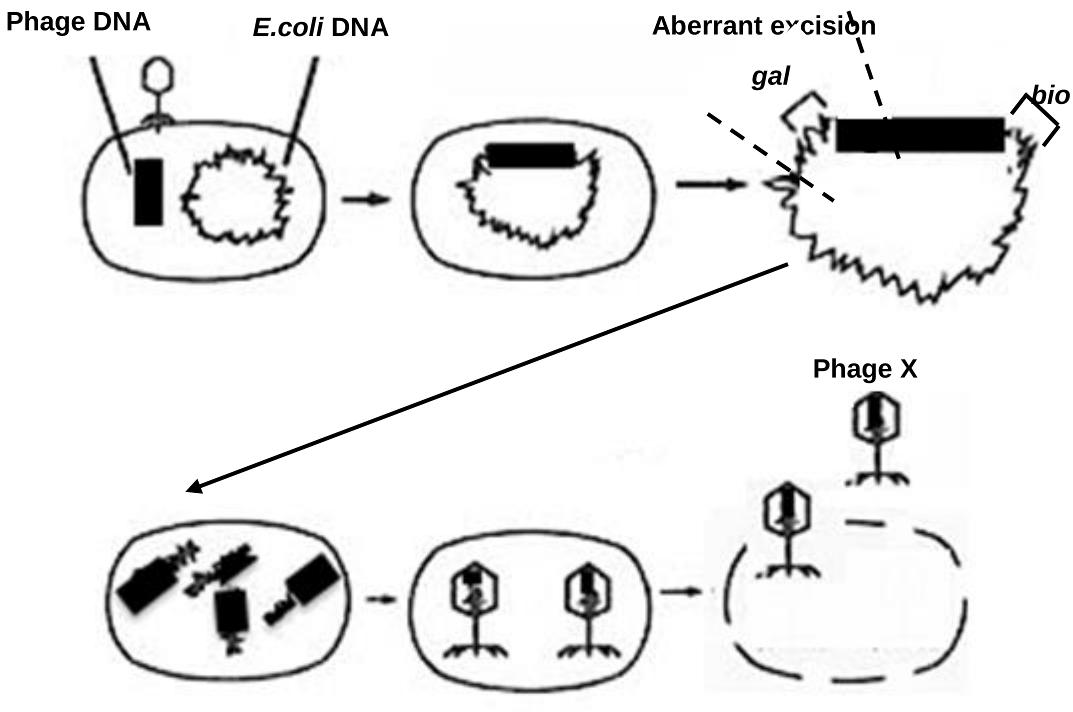

Image adapted from http://medchrome.com

Fig. 2.2

2. **(c)** Compare how lytic cycle differs from lysogenic cycle. [3]

(d) With reference to **Fig. 2.2**, suggest why the mutated strain of *E.coli* can regain its cell wall synthesis function if it is infected by Phage **X**. [2]

[Total: 12]

PART I

3. **Fig. 3.1** is an electron micrograph of a cell.

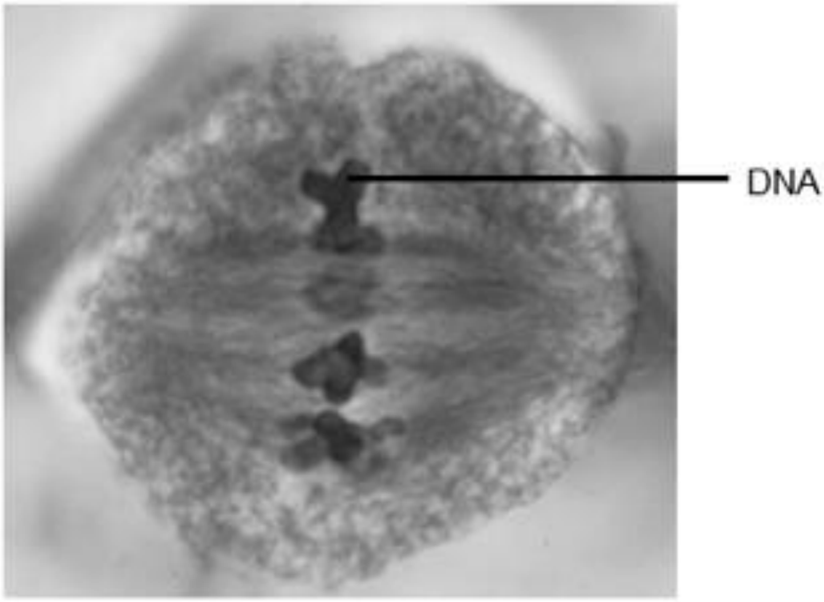

Image adapted from Carolina Biological Supply Company

Fig. 3.1

(a) Describe how the DNA shown in **Fig. 3.1** is folded and the resultant impact on its transcriptional activity. [3]

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(b) Describe how else transcriptional activity of a cell can be prevented. [2]

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PART II

3. Cells rely on chaperone proteins to prevent nearby proteins from interfering with protein folding. Protein folding can occur in the cytoplasm and proteins must fold into defined three-dimensional structures to gain functional activity. As protein molecules are highly dynamic, constant chaperone surveillance is required to ensure proper functioning of proteins.

(c) Explain how chaperone proteins help in the regulation of gene expression. [3]

(d) Suggest the advantage of regulating gene expression at this level. [1]

(e) Describe how gene expression can be down-regulated at the protein level. [3]

[Total: 12]

4. **PART I**

(a) Explain what is meant by the following terms:

(i) heterozygous [1]

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(ii) genotype [1]

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Rickets is a childhood disorder involving the softening and weakening of bones. It is usually caused by a lack of vitamin D, calcium ions or phosphate ions. A rare form of rickets that cannot be successfully treated with vitamin D therapy is caused by a mutant allele on the X chromosome.

Fig. 4.1 shows a pedigree for a family that has a history of this condition.

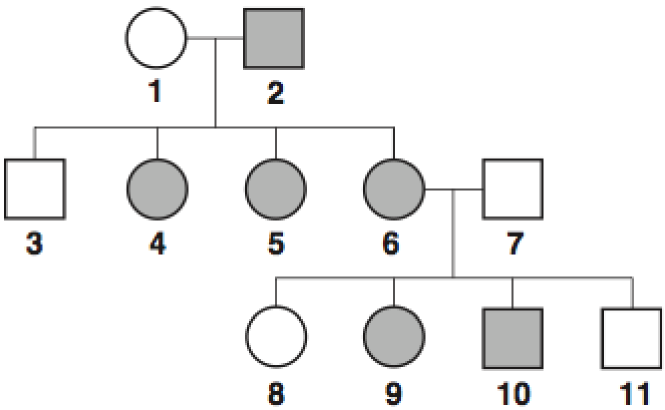

Fig. 4.1

(b) Using the symbols,

X^R for the mutant allele on the X chromosome;

X^r for the non-mutant allele on the X chromosome,

state the genotypes of the following individuals. [3]

(i) Individual 1: ______________________

(ii) Individual 3: ______________________

(iii) Individual 9: ______________________

4. **PART II**

The budgerigar, *Melopsittacus udulatus*, shown in **Fig. 4.2**, is a parrot that is native to Australia.

Image adapted from University of Cambridge International Examinations, 2012

Fig. 4.2

A budgerigar can have blue, green, yellow or white feathers. Two genes, A/a and D/d, are involved in the inheritance of feather colour in budgerigars.

- A bird which has at least one dominant allele **A** but is homozygous for **d** has blue feathers.
- A bird which has at least one dominant allele **D** but is homozygous for **a** has yellow feathers.
- A bird with at least one dominant allele **A** and one dominant **D** allele has green feathers.
- A bird that is homozygous for **a** and **d** has white feathers.

(c) A student crossed two green-feathered budgerigars.

Draw a genetic diagram of this cross to show the probability of producing offspring with yellow feathers is 25%. [5]

4. **(d)** Explain how the two genes interact to give rise to the different feather colours. [2]

[Total: 12]

5. Cathy investigated the pathway by which carbon dioxide is converted to organic compounds during photosynthesis. She used the apparatus shown in **Fig. 5.1**.

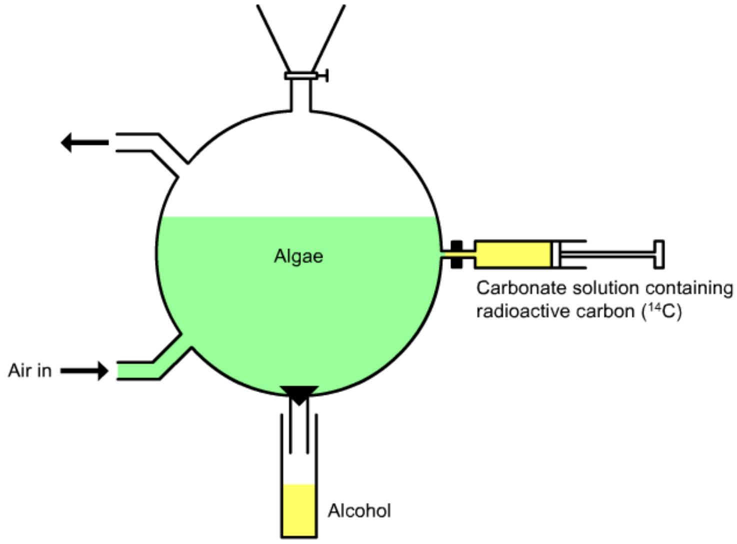

Image adapted from http://www.snabonline.com

Fig. 5.1

While the apparatus was in the dark room, Cathy supplied the algal cells with $^{14}CO_2$. The contents of the apparatus were thoroughly mixed and light was switched on subsequently. At five-second intervals, she released a few of the cells into hot alcohol, which killed the cells very quickly. The intermediates of the reactions were subsequently analysed and the results were shown in **Fig. 5.2**.

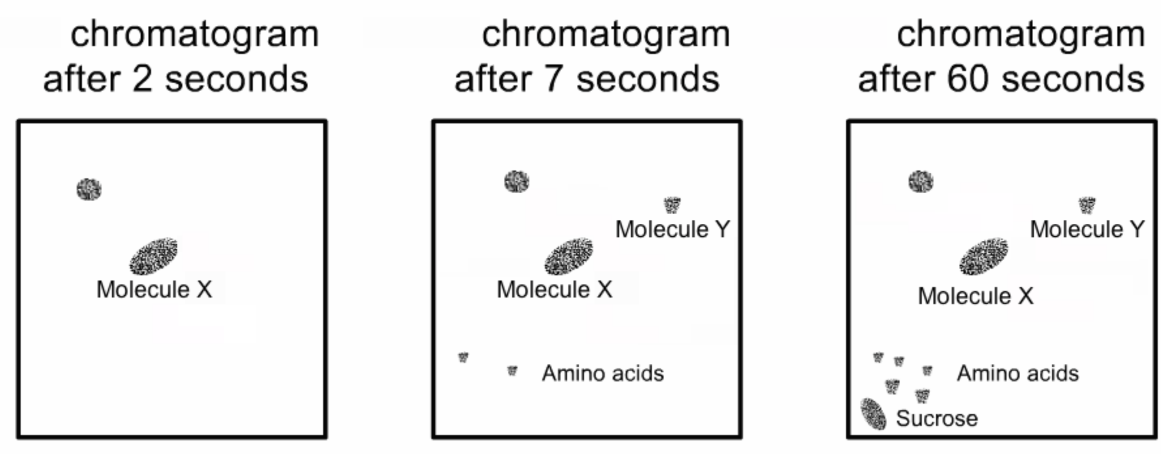

Image adapted from http://www.snabonline.com

Fig. 5.2

5. **(a)** Describe how Molecule **X** and Molecule **Y** are formed. [4]

Dinitrophenol is a metabolic poison that can embed within the thylakoid membranes of chloroplasts and provide an alternate route for H^+ to diffuse across the thylakoid membranes.

(b) Explain how the concentration of intermediates of Calvin cycle is affected by dinitrophenol. [2]

5. A further experiment was carried out by another student to determine the concentration of carbon dioxide in the leaves of plants at different times of the day. The results are shown in **Table 5.1**.

Table 5.1

Mean carbon dioxide concentration (ppm)	
8pm to 4am	8 am to 4pm
328	106

(c) Explain how Calvin cycle and Krebs cycle can account for the difference in concentration of carbon dioxide in the leaves for the two periods shown in **Table 5.1**. [4]

(d) Explain why higher concentration of carbon dioxide would increase carbon fixation during the period 8am to 4pm. [2]

[Total: 12]

6. **PART I**

In recent years, significant advances have been made in defining the regulatory pathways that control normal and abnormal cardiac valve development. One of such includes the receptor tyrosine kinase activation and downstream signaling pathway shown in **Fig. 6.1**. The epidermal growth factor (EGF) activates the MAPK/ERK pathway to induce the transcription of targeted gene.

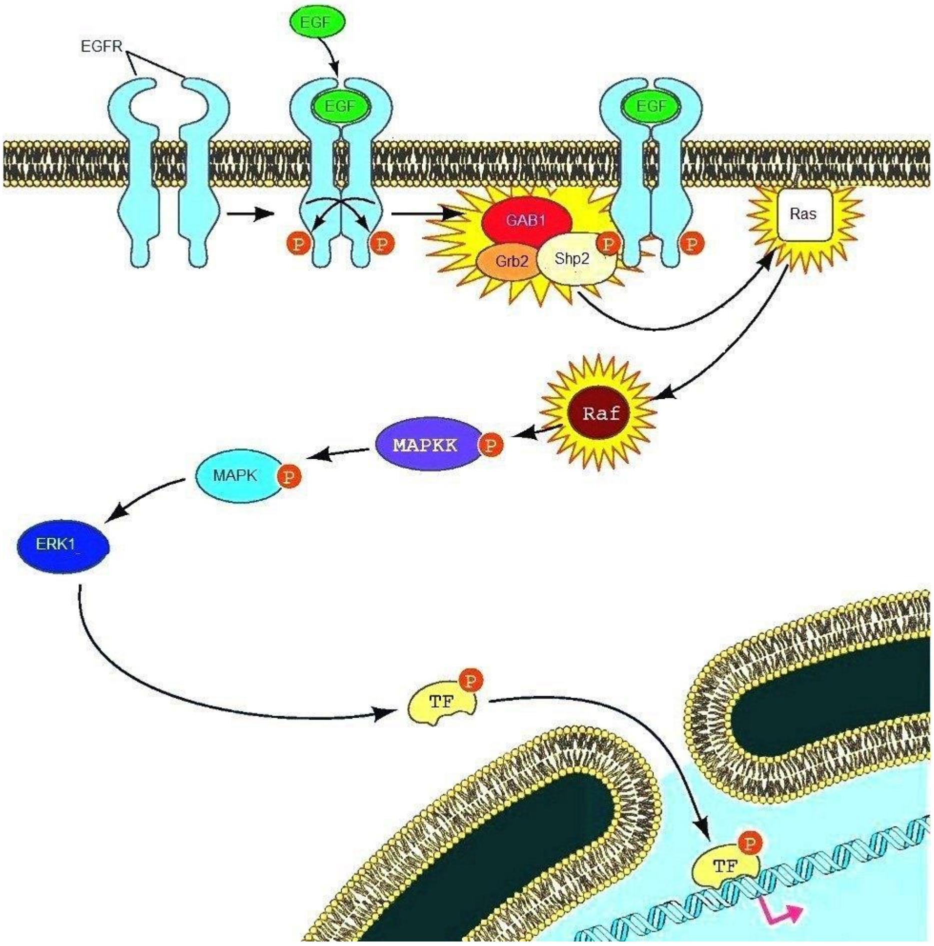

Image adapted from http://physiologyonline.physiology.org/content/20/6/390/F1

Fig. 6.1

(a) With reference to **Fig. 6.1**,

(i) describe how EGFR is activated, [3]

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6. **(a)** **(ii)** describe how the signal from the activated receptor is transduced to Raf. [3]

6. **PART II**

Oral glucose tolerance test is often employed to diagnose diabetes when simple blood and urine tests give inconclusive results. Following a 12 hour fast, a blood sample is then taken to measure the patient's fasting blood glucose level. He is then given a drink containing 75g of glucose, and his blood glucose concentration tracked for the next 2 hours.

Fig. 6.2 shows the blood glucose concentration of two individuals, one healthy and one with diabetes.

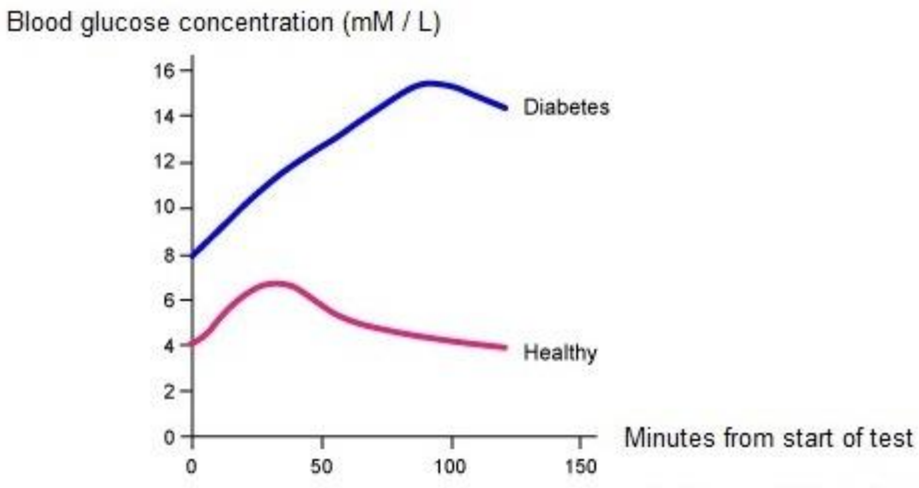

Image adapted from ABPI schools

Fig. 6.2

(b) Describe and explain the results of the test for the diabetic patient in the first 100 minutes. [3]

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(c) Suggest a reason for the drop in blood glucose concentration for the diabetic patient. [1]

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[Total: 10]

7. Many species of anole lizards are found throughout the Caribbean and the surrounding mainland. Each species is found only on one island or a small group of islands, apart from *Anolis carolinensis* which is found in mainland Florida.

Fig. 7.1 shows the distribution of four species of anole lizards.

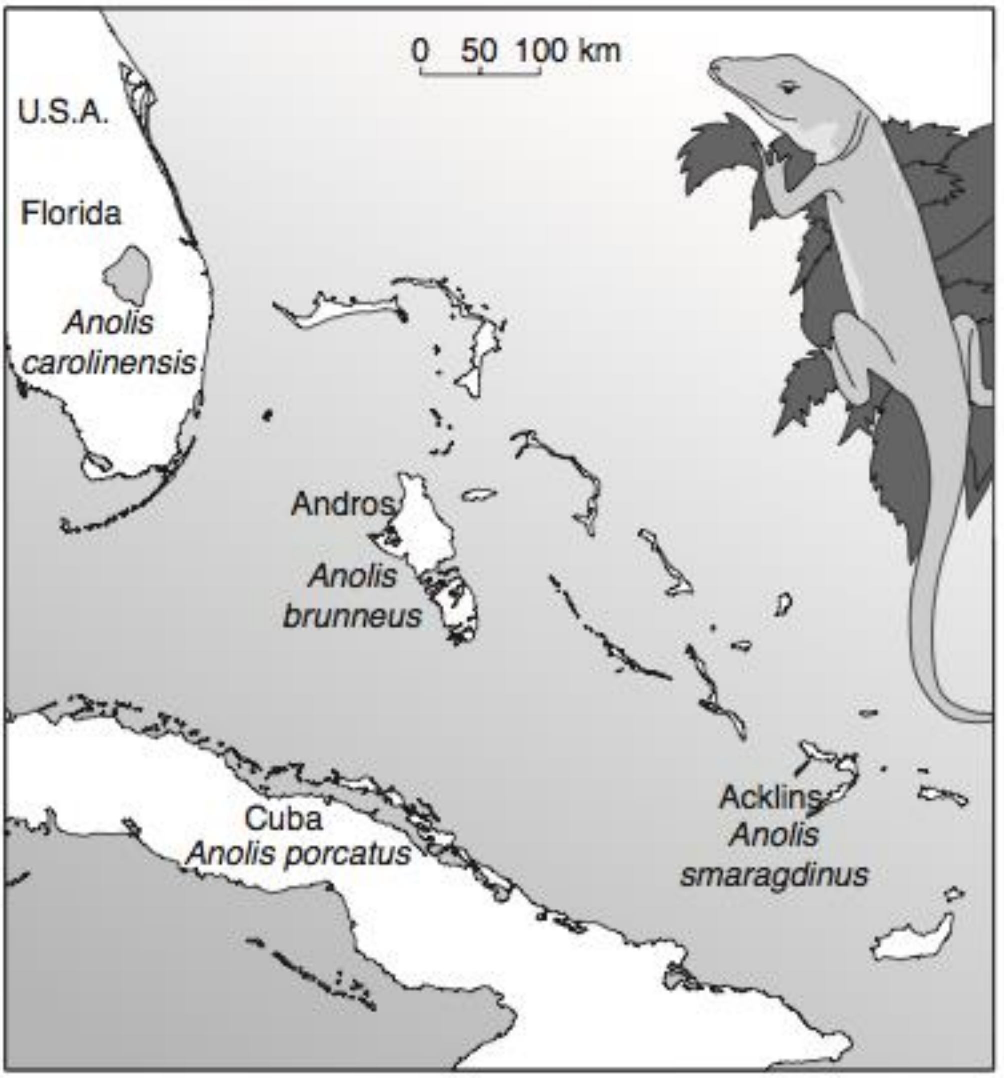

Image adapted from University of Cambridge International Examinations, 2010

Fig. 7.1

A student carried out DNA analysis to investigate the relationships between these four lizards species. The base sequences of a region of mitochondrial DNA (mtDNA) from the four species were compared. **Table 7.1** shows the difference in base sequence of the species.

7. **Table 7.1**

	A. brunneus			
A. brunneus		***A. smaragdinus***		
A. smaragdinus	12.1		***A. carolinensis***	
A. carolinensis	16.7	15.0		***A. porcatus***
A. porcatus	11.3	8.9	13.2	

The student put forward the hypothesis that the three species, *A. brunneus*, *A. smaragdinus* and *A. carolinensis*, have originated from three separate events in which a few individuals of *A. porcatus* spread directly from Cuba to three different places.

(a) Suggest why changes in mtDNA can be used to support the neutral theory of molecular evolution in closely related species. [3]

(b) Explain how the results in **Table 7.1** support their hypothesis. [3]

7. **(c)** Suggest why the mtDNA of *A. porcatus* has the least differences with *A. smaragdinus*, compared to *A. brunneus* and *A. carolinensis*. [1]

(d) Explain how a population of *A. porcatus* that became isolated on an island could evolve into a new species. [5]

[Total: 12]

<u>**Section B (20 marks)**</u>

Answer **one** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

A **NIL** return is necessary if you have not attempted this section.

8. **(a)** Compare cellulose and collagen. [8]

(b) Explain why animals store triglycerides in addition to glycogen. [5]

(c) Relate the structure of mitochondria to its role in oxidation of glucose. [7]

[Total: 20]

9. **(a)** Describe how the structure of a cholinergic synapse facilitates impulse transmission. [9]

(b) Explain the cause of refractory period and its significance to impulse transmission. [5]

(c) Outline the function of cell membranes. [6]

[Total: 20]